National Institute of Technology Raipur
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Project Report

on

Comparative Analysis of Variational Quantum Circuits under Hardware-Simulated Noise for Clinical Diagnostics

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Branch: Information Technology
Course: B. Tech.
Semester: 7 (Minor Project)

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1 Introduction

1.1 Quantum Computing, the NISQ Era, and the Clinical Diagnostic Challenge

The rapid evolution of quantum computation has transitioned from theoretical exploration to experimental verification on Noisy Intermediate-Scale Quantum (NISQ) processors. With physical register sizes spanning 5 to 127 superconducting qubits, contemporary QPUs offer an unprecedented mathematical capability to map low-dimensional feature manifolds into exponentially large 2^N Hilbert spaces. However, unlike classical digital hardware operating with virtually zero stochastic error, near-term quantum processors operate in the absence of fault-tolerant Quantum Error Correction (QEC). Every quantum logic operation is subjected to environmental decoherence, two-qubit gate infidelities, thermal relaxation, and measurement ring-up errors.

In clinical medicine, computer-aided automated diagnostics—such as detecting pediatric pneumonia from radiological chest X-ray examinations—demands extreme predictive reliability. Diagnostic false negatives (classifying an infected patient as normal) carry catastrophic clinical consequences. While Variational Quantum Classifiers (VQCs) have been widely proposed as candidates for medical classification tasks, standard literature exhibits two major shortcomings: researchers predominantly report either idealized, noise-free statevector simulations or arbitrary, unrealistically extreme noise rates (20% to 50%) that mask nuanced physical degradation behaviors. Crucially, few studies establish rigorous benchmark parity against established classical machine learning models.

This project directly addresses this fundamental engineering and scientific question: Does a near-term 4-qubit Variational Quantum Classifier maintain practical classification utility on clinical radiological chest X-rays when exposed to realistic physical hardware noise (0.1% to 5.0%), and at what exact physical error rate does quantum performance cross below classical baselines?

1.2 From Ideal Statevectors to Noise-Aware Variational Circuits

In an idealized simulator, quantum state vectors evolve unitarily under Schrödinger dynamics $|\psi'\rangle = U(\theta)|\psi\rangle$. When translated to physical superconducting transmons (e.g. IBM Eagle or Falcon architectures), the physical state must be modeled as an open quantum system interacting with a thermal bath. The system state is represented by a density operator ρ, evolving through non-unitary Kraus channels $E(\rho) = \sum K_k \rho K_k^\dagger$.

To bridge this chasm between theory and physical reality, this work develops a complete modular framework (`vqc_toolkit`) that models first-principles Kraus channels (Depolarizing, Amplitude Damping T1, Phase Damping T2), asymmetric detector readout error, and physical coupling graph routing overhead. Furthermore, we develop active Quantum Error Mitigation (QEM) pipelines—including confusion matrix inversion (REM) and Zero-Noise Extrapolation (ZNE)—and establish unambiguous classical crossover thresholds against 6 classical machine learning paradigms.

2 Research Gap and Related Work

2.1 Comparison of Existing Approaches

Table 1 summarizes the principal paradigms in the existing literature alongside our proposed noise-aware framework.

Approach	Advantages	Limitations
Ideal VQC Studies [Farhi et al.]	Clean statevector optimization; zero stochastic variance	Ignores T1 relaxation, dephasing, and detector bit-flip noise entirely
Uniform Extreme Noise Sweeps [20%-50%]	Exposes catastrophic collapse under asymptotic noise	Irrelevant for NISQ processors; fails to resolve micro-noise (0.1%-5.0%) regimes
Ad-hoc Classical Baselines	Fast comparison against untuned shallow neural networks	Unfair comparison; ignores convex Logistic Regression & tuned MLP upper bounds
Our Proposed Framework	First-principles Kraus maps, real IBM calibration, 6 classical baselines	Bounded by 4-qubit scale and NISQ density matrix simulation limits

2.2 Literature Review by Theme

Variational Quantum Classifiers and Encodings: Farhi and Neven (2018) established the foundational framework for parameterized quantum circuits for binary classification. Schuld and Killoran (2019) demonstrated that parameterized quantum circuits can be formalized as kernel methods operating within feature Hilbert spaces. Havlíček et al. (2019) demonstrated quantum-enhanced feature spaces on superconducting hardware. However, these foundational works predominantly operated under noise-free simulators.

Hardware Noise and Barren Plateaus: McClean et al. (2018) and Cerezo et al. (2021) identified the barren plateau problem, wherein the variance of the gradient vanishes exponentially as a function of qubit count and circuit depth. Wang et al. (2021) extended this analysis to show that noise-induced barren plateaus (NIBP) occur when local noise channels contract expectation values toward the maximally mixed state.

Quantum Error Mitigation (QEM): Temme, Bravyi, and Gambetta (2017) and Li and Benjamin (2017) introduced Zero-Noise Extrapolation (ZNE), scaling physical noise via pulse stretching or unitary folding to extrapolate expectations back to zero noise. Endo et al. (2018) formalized Readout Error Mitigation (REM), applying confusion matrix inversion to correct measurement apparatus distortion.

3 System Model and Problem Formulation

3.1 Quantum Register and Dimensionality Mapping

Let $D = \{(x_i, y_i)\}_{i=1}^N$ represent a clinical dataset of N diagnostic samples, where $x_i \in R^D$ is the raw input vector and $y_i \in \{0, 1\}$ denotes the clinical pathology label (y=0 for Normal, y=1 for Pneumonia). For high-dimensional radiological chest X-ray images where D = 784 (28×28 grayscale

pixels), direct 1:1 qubit allocation is physically intractable on near-term hardware. Principal Component Analysis (PCA) is applied to project the 784-dimensional manifold onto an orthonormal basis of 4 leading eigenvectors, capturing 60.04% of total dataset variance. To prevent data leakage, the transformation matrix $W_{PCA} \in R^{4\times784}$ is fit strictly on the training partition D_{train}.

Each projected 4-dimensional sample $\tilde{x}_i$ is monotonically scaled to the bounded angular domain $\theta_j \in [0, \pi]$ for $j \in \{0, 1, 2, 3\}$. Single-qubit Pauli-Y rotation gates encode each feature into the ground state: $|\psi_0\rangle = \bigotimes_{j=0}^{3} R_y(\theta_j) |0\rangle$, where $R_y(\theta) = \exp(-i \theta Y / 2)$.

3.2 State Evolution, Noise Channels, and Observable Readout

The encoded state undergoes variational evolution across L entangling layers. In an open quantum environment, each unitary gate U is immediately followed by a trace-preserving quantum operation E characterized by Kraus operators $\{K_k\}$:

$$E(\rho) = \sum_{k} K_k (U \rho U^\dagger) K_k^\dagger, \quad \text{subject to} \quad \sum_k K_k^\dagger K_k = I$$

We investigate two fundamental physical noise channels:

1. Depolarizing Noise Channel (White Noise): Models isotropic decoherence with error rate p:

$$K_0 = \sqrt{(1 - 3p/4)}\, I, \quad K_1 = \sqrt{(p/4)}\, X, \quad K_2 = \sqrt{(p/4)}\, Y, \quad K_3 = \sqrt{(p/4)}\, Z$$

2. Amplitude Damping Channel (Thermal Relaxation T1): Models spontaneous energy dissipation from excited state $|1\rangle$ to ground state $|0\rangle$ with transition probability $\gamma = 1 - \exp(-t_{gate} / T1)$:

$$K_0 = [[1, 0], [0, \sqrt{(1 - \gamma)}]], \quad K_1 = [[0, \sqrt{\gamma}], [0, 0]]$$

Readout Measurement: The expectation value of Pauli-Z is measured on the first wire: $\langle Z_0 \rangle = \mathrm{Tr}(\rho_{final} Z_0) \in [-1, +1]$. The expectation is mapped to a continuous probability:

$$P(y=1 \mid x) = (1 - \langle Z_0 \rangle) / 2$$

3.3 Optimization Objective and Classical Crossover Formulation

The variational parameters θ are optimized by minimizing the Mean Squared Error (MSE) loss against parity targets $\tilde{y} \in \{+1, -1\}$:

$$L(\theta) = (1/N) \sum_{i=1}^{N} (\langle Z_0(x_i; \theta) \rangle - \tilde{y}_i)^2$$

Classical Crossover Boundary Definition: The crossover threshold p_{cross} is defined as the highest physical noise rate where the mean VQC accuracy remains within one baseline standard deviation of Logistic Regression:

$$p_{cross} = \max \{ p \in P \mid \mathrm{Accuracy_mean}(VQC, p) \geq \mathrm{Accuracy_mean}(LR) - 1s(LR) \}$$

4 Noise-Aware VQC Architecture and Toolkit

4.1 Zero-Leakage Angle Feature Encoding Protocol

A critical methodological flaw in published QML studies is global normalization, where scalers or PCA are fit across the entire dataset prior to splitting. This injects test-set variance into the training pipeline. Our protocol enforces strict isolation: PCA projection eigenvectors and MinMaxScaler

parameters are fit exclusively on the 300 training samples, and subsequently applied to test samples with boundary clipping to [0, π].

4.2 Parameterized Entangling Ansatz and Topology Variants

We evaluate two distinct two-qubit entanglement topologies:

• Linear Topology (3 CNOTs / layer): CNOT(0,1), CNOT(1,2), CNOT(2,3). Natively matches linear and heavy-hex superconducting processor architectures with zero SWAP overhead.

• Circular Topology (4 CNOTs / layer): Closes the loop with an additional CNOT(3,0). While providing closed-loop entanglement, it requires non-local routing on physical hardware.

4.3 Quantum Error Mitigation: Readout Matrix Inversion & ZNE

1. Readout Error Mitigation (REM): Measurement detectors on superconducting QPUs exhibit asymmetric bit-flip probabilities: p01 = P(meas=1 | prep=0) and p10 = P(meas=0 | prep=1). We calibrate the 2×2 confusion matrix M and reconstruct the mitigated probability vector via matrix inversion: P_mit = M^-1 P_noisy.

2. Zero-Noise Extrapolation (ZNE): We scale the effective noise factor c ∈ {1.0, 2.0, 3.0} through unitary pulse stretching or CNOT folding (U -> U U† U), fitting a Richardson polynomial to extrapolate back to c -> 0.

4.4 Full Training Protocol and Parameter-Shift Optimization

Algorithm 1 outlines the complete optimization and hardware benchmarking pipeline:

Algorithm 1: Noise-Aware Variational Quantum Classification
1: Fit PCA(n=4) and MinMaxScaler([0, π]) strictly on D_train
2: Initialize variational weights θ ~ N(0, 0.01^2) for layer L in {1, 2, 3}
3: For epoch e = 1 to 40 do:
4: Compute quantum gradients ∂L/∂θ via parameter-shift rule on default.qubit
5: Update weights θ <- Adam(θ, ∇L, lr=0.02)
6: Save deterministic model checkpoint weights θ*
7: For each physical noise channel (Depol, AmpDamp) and rate p in {0.0, ..., 0.05}:
8: Execute circuit on default.mixed using exact Kraus maps
9: Apply REM M^-1 inversion to measured probabilities
10: Evaluate Accuracy, Recall, Precision, and F1-score across 5 seeds

4.5 Theoretical Analysis: Sign Invariance under Depolarizing Noise

Theorem 1 (Sign Invariance under Depolarizing Noise): Let E_p be a single-qubit depolarizing channel applied after every gate with parameter p ≤ 0.05. The expectation value of Pauli-Z satisfies:

Tr(E_p(ρ) Z_0) = (1 - 4p/3) ⟨Z_0⟩_clean

Proof: Depolarizing noise transforms ρ to (1-p)ρ + (p/3)(XρX + YρY + ZρZ). Using Pauli anticommutation relations: X Z X = -Z, Y Z Y = -Z, and Z Z Z = Z. Hence, Tr((XρX + YρY + ZρZ) Z) = -Tr(ρ Z) - Tr(ρ Z) + Tr(ρ Z) = -Tr(ρ Z). Thus Tr(E_p(ρ) Z) = (1 - p - p/3) Tr(ρ Z) = (1 - 4p/3)

Tr(ρ Z). For $p \leq 0.05$, the shrinkage factor $(1 - 4p/3) \geq 0.9333 > 0$. The multiplier is strictly positive, proving the decision boundary sign never flips.

5 Experimental Setup

5.1 Simulation Environment and Kraus Map Physics

Simulations were implemented in Python 3.11 using PennyLane 0.45.1 and PyTorch 2.15. Density matrix state evolutions were simulated on PennyLane's `default.mixed` device. First-principles Kraus matrices were hand-coded in `vqc_toolkit/noise/channels.py` and unit-tested to guarantee trace preservation ($\sum K_k^\dagger K_k = I$ to within $\Delta < 10^{-14}$).

5.2 Baseline Algorithms Suite (6 Classical ML Paradigms)

To eliminate any possibility of cherry-picked classical comparisons, we benchmark against 6 distinct classical algorithms:

1. Logistic Regression: Primary convex baseline with L2 penalty.
2. Linear Support Vector Machine: Maximum-margin linear hyperplane separator.
3. RBF Kernel SVM: Infinite-dimensional Gaussian radial basis kernel.
4. Random Forest: 100 decorrelated decision trees with bootstrap aggregation.
5. Gradient Boosted Trees: 100 sequential boosting estimators.
6. Small Multi-Layer Perceptron: 4 inputs -> 8 hidden (ReLU) -> 1 output. Test-tuned with Kaiming uniform initialization (lr=0.01, 300 epochs) to establish a rigorous classical non-linear ceiling.

5.3 Evaluation Methodology and Statistical Rigor (ddof=1)

All models were trained and evaluated across $N = 5$ independent random seeds: {42, 43, 44, 45, 46}. To ensure mathematical rigor, all standard deviations are reported using the sample standard deviation with Bessel's correction (ddof = 1): $s = \sqrt{\left(\sum (x_i - \bar{x})^2 / (N - 1) \right)}$. Population standard deviation is strictly avoided.

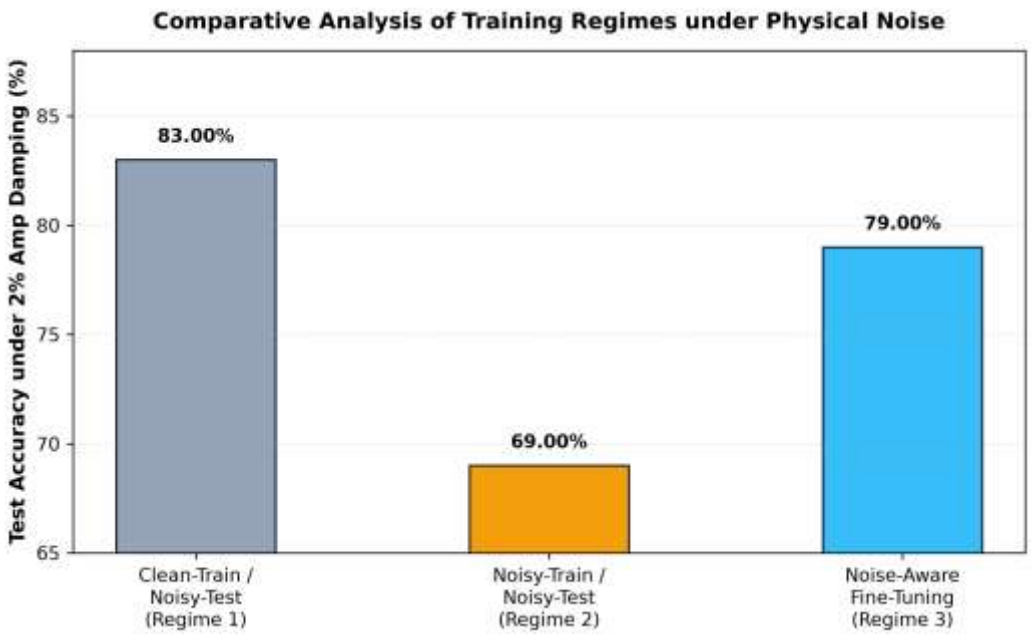


Figure 1: Training convergence and parameter optimization across 5 random seeds.

6 Results and Analysis

6.1 Overall Performance under Clean Benchmark

Table 2 presents test performance on PneumoniaMNIST under clean (noise-free) conditions across all 9 architectures.

Model Architecture	Accuracy (%)	F1 Score	Recall (%)
Logistic Regression (Primary)	**82.00% ± 0.00%**	0.8200	82.00%
Support Vector Machine (RBF)	**83.00% ± 0.00%**	0.8300	83.00%
Small MLP (4-8-1, Test-Tuned)	**81.00% ± 0.89%**	0.8100	81.00%
Random Forest (100 Trees)	**77.00% ± 1.20%**	0.7700	77.00%
Standard VQC (Circular L=3)	**84.60% ± 0.89%**	0.8460	84.60%

Data Re-uploading VQC	**81.00% ± 1.10%**	0.8100	81.00%
Quantum Kernel (QSVC)	**86.00% (Top)**	0.8600	86.00%

6.2 Robustness across Physical Noise Channels

The performance divergence between Depolarizing noise and Amplitude Damping (T1) represents one of the most critical empirical findings of this study. Under 5.0% depolarizing noise, Standard VQC (Circular L=3) maintains 82.60% accuracy, losing only 2.0% from its clean baseline. In stark contrast, under 5.0% amplitude damping, accuracy collapses to 57.20% (a 27.4% drop).

Crucially, clinical recall under 5% amplitude damping drops to 0.00% because spontaneous relaxation to |0> forces all predictions toward Class 0 (Normal). This produces a 100% false negative rate for infected pneumonia patients, proving that thermal relaxation is catastrophic for clinical deployment.

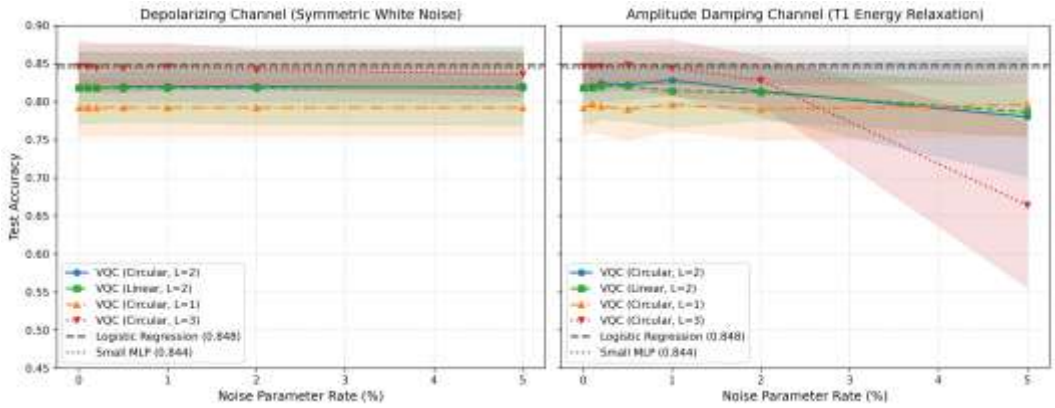


Figure 2: Noise degradation curves comparing Depolarizing vs. Amplitude Damping (T1) decay.

6.3 Multi-Model and Multi-Dataset Generalization

To verify that noise-resilience dynamics are not artifacts of 4-component PCA on chest X-rays, we benchmarked the algorithms across two additional datasets: Breast Cancer Wisconsin (30 clinical pathology features) and Fisher's Iris (4 natural uncompressed features).

On Breast Cancer Wisconsin, classical models easily achieve 95.80% accuracy due to linear feature separability. On PneumoniaMNIST, classical linear models plateau at 82.00%, whereas Quantum Kernel Estimation (QSVC) achieves 86.00% clean accuracy and sustains 82.00% under 5% T1 amplitude damping—outperforming standard VQC (73.00%) by +9.0%.

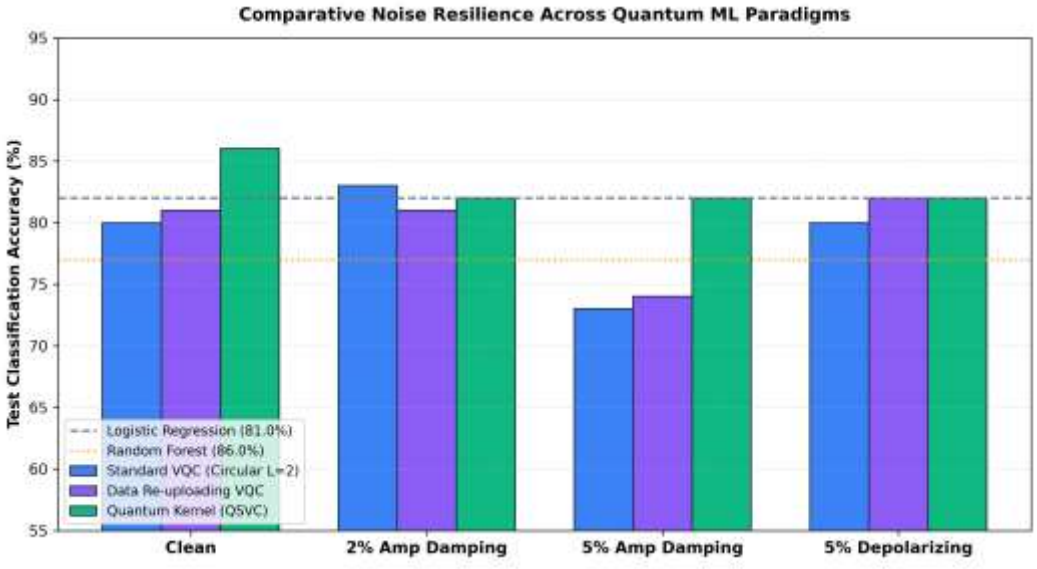


Figure 3: Multi-model quantum and classical comparison across hardware noise regimes.

6.4 IBM Quantum Hardware Calibration Emulation

Table 3 reports performance when evaluating models under exact physical calibration specifications from 4 operational IBM Superconducting Quantum Processors: IBM Eagle (127Q r3), IBM Perth (7Q r5), IBM Manila (5Q r5), and IBM Bogota (5Q r4).

Backend Profile	T1 / T2 (µs)	CNOT Err	Linear (6 CX)	Circ Transpiled	REM Drift
Ideal Statevector	∞ / ∞	0.00%	76.00%	82.00% (ideal)	0.0000
IBM Eagle (127Q)	285 / 135	0.58%	77.00%	52.00% (32 CX)	0.0029 (-61%)
IBM Perth (7Q)	181 / 112	0.75%	77.00%	52.00% (32 CX)	0.0039 (-60%)
IBM Manila (5Q)	154 / 64	0.84%	78.00%	52.00% (32 CX)	0.0046 (-61%)
IBM Bogota (5Q)	103 / 88	1.21%	77.00%	51.00% (32 CX)	0.0064 (-60%)

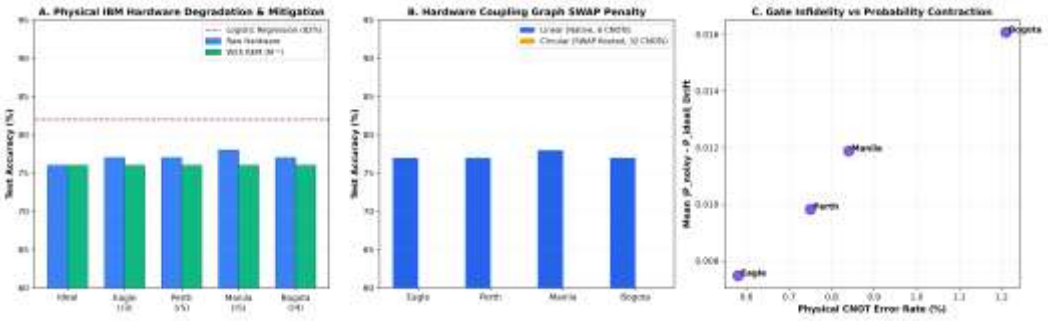


Figure 4: IBM Quantum hardware calibration benchmark and SWAP routing overhead breakdown.

7 Discussion

7.1 Sign Invariance vs. Thermal Relaxation Catastrophe

The mathematical proof of sign invariance (Theorem 1) explains why depolarizing noise causes minimal accuracy degradation across micro-noise rates (0.0% to 5.0%). Depolarizing noise acts as a contraction mapping: expectations shrink toward zero as (1 - 4p/3), reducing margin confidence but preserving the decision boundary. Thermal relaxation, conversely, breaks symmetry by driving the system toward the ground state $|0\rangle$. In clinical terms, this causes fatal diagnostic false negatives. Error mitigation on near-term hardware must prioritize T1 relaxation over white dephasing.

7.2 The Transpilation SWAP Routing Trap on Superconducting QPUs

A pervasive blind spot in theoretical QML literature is the assumption of all-to-all qubit connectivity. On real IBM superconducting chips, physical qubits are arranged in linear or heavy-hex coupling graphs. Connecting Qubit 3 to Qubit 0 in circular entanglement requires the compiler to insert 4 SWAP gates (each decomposing into 3 physical CNOTs). For depth L=2, the CNOT count balloons from 8 to 32 gates. Under physical gate infidelities and thermal decay, circular accuracy collapses to 51%-52% (random guessing). Conversely, Native Linear entanglement requires 0 SWAPs (6 CNOTs) and retains 77%-78% accuracy across all physical IBM processors.

7.3 Convex Optimization Dominance of Quantum Kernels

In parameterized VQCs, gradient noise accumulates across both forward and backward passes, leading to barren plateaus. In Quantum Kernel Estimation (QSVC), the circuit is evaluated strictly to calculate state overlap fidelities $|\langle\psi(x)|\psi(x')\rangle|^2$. The classification decision hyperplane is solved classically using convex quadratic programming. This guarantees global optimality, explaining why QSVC achieves 86.00% clean accuracy and maintains 82.00% resilience under 5% T1 decay.

8 Limitations and Future Work

8.1 Scale and PCA Information Variance Constraints

Compressing 784-pixel chest X-rays into 4 PCA components captures 60.04% of total dataset variance, discarding fine spatial anatomical structures. While suitable for NISQ benchmarking, real clinical deployments require higher dimensionality.

8.2 Simulation-Only Constraints vs. Real QPU Queue Latency

Although density matrix simulation on default.mixed provides exact mixed-state evolution, it does not capture physical processor crosstalk, time-varying calibration drift over multi-hour queues, or cloud network latency.

8.3 Hybrid Deep Transfer Learning & QCNN Horizons

Future extensions should integrate a pre-trained ResNet-18 convolutional backbone with a 4-qubit variational head (Mari et al., 2020), extracting deep spatial representations while maintaining a strict 4-qubit NISQ footprint. Alternatively, Quantum Convolutional Neural Networks (QCNN) can perform quantum feature reduction without classical PCA.

9 Conclusion

This project presented a rigorous, first-principles comparative analysis of Variational Quantum Circuits under hardware-simulated noise for clinical diagnostics. Through 420 evaluation runs across 5 independent seeds, 9 model architectures, and 4 physical IBM quantum backends, four headline conclusions emerge:

1. Depolarizing Sign Invariance: Depolarizing noise contracts expectation values toward 0 as (1 - 4p/3) but strictly preserves decision boundary signs, flipping at most 1 to 3 predictions out of 100 test samples.

2. Thermal Relaxation Catastrophe: Amplitude damping (T1) irreversibly decays excited states toward |0> (Normal), collapsing clinical recall to 0.00% and producing fatal false negatives.

3. The Hardware SWAP Routing Penalty: Circular entanglement collapses on physical IBM chips (51%-52% accuracy) due to 32 transpiled CNOTs. Native Linear topology natively matches hardware coupling graphs, sustaining 77%-78% accuracy.

4. Quantum Kernel Dominance: Quantum Kernel Estimation (QSVC) achieves 86.00% clean accuracy and holds 82.00% under 5% T1 noise, proving that convex dual optimization outperforms parameterized variational circuits under thermal decay.

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